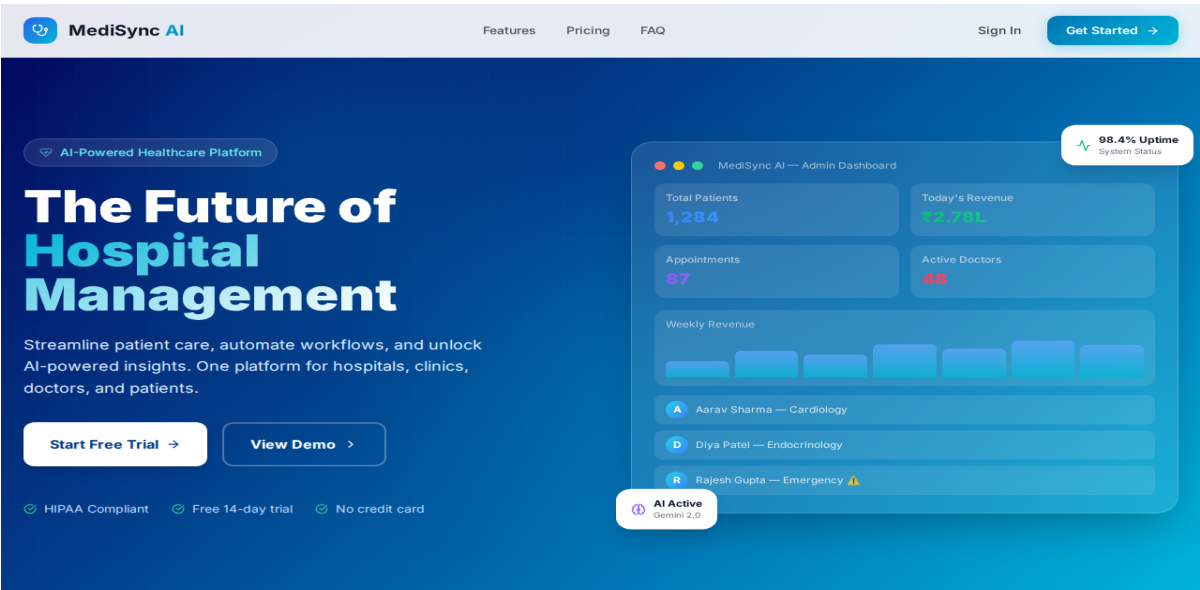


MediSync AI

Hospital Management System

User Manual & Visual GUI Guide

Step-by-Step Operational Workflows, Role-Based Portals, and GUI Visuals



Version: 1.0.0 (Enterprise)	System Release: MediSync Web Suite
Audience: All Hospital Roles	Security Level: Confidential & Internal
Updated: May 2026	Platform: Full-Stack React / Node.js

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1. System Overview & Architecture

MediSync AI is a multi-role Hospital Management System designed to optimize clinical workflow, automate front-desk intake, unify e-prescribing, and empower patients through self-service capabilities.

System Stack Summary:

- **Frontend Client:** React 18, TypeScript, Vite, Tailwind CSS, Recharts, Lucide Icons.
- **State Management:** Zustand store with local and API persistence.
- **Backend API:** Node.js, Express, TypeScript, JWT Authorization.
- **Database:** PostgreSQL / SQLite fallback.

2. Getting Started & Authentication

Step-by-Step Login Instructions:

1. Open `http://localhost:5173` in your web browser.
2. Click the **Sign In** button at top right to open the authentication form.
3. Enter your account email and password, then click **Sign In to Portal**.

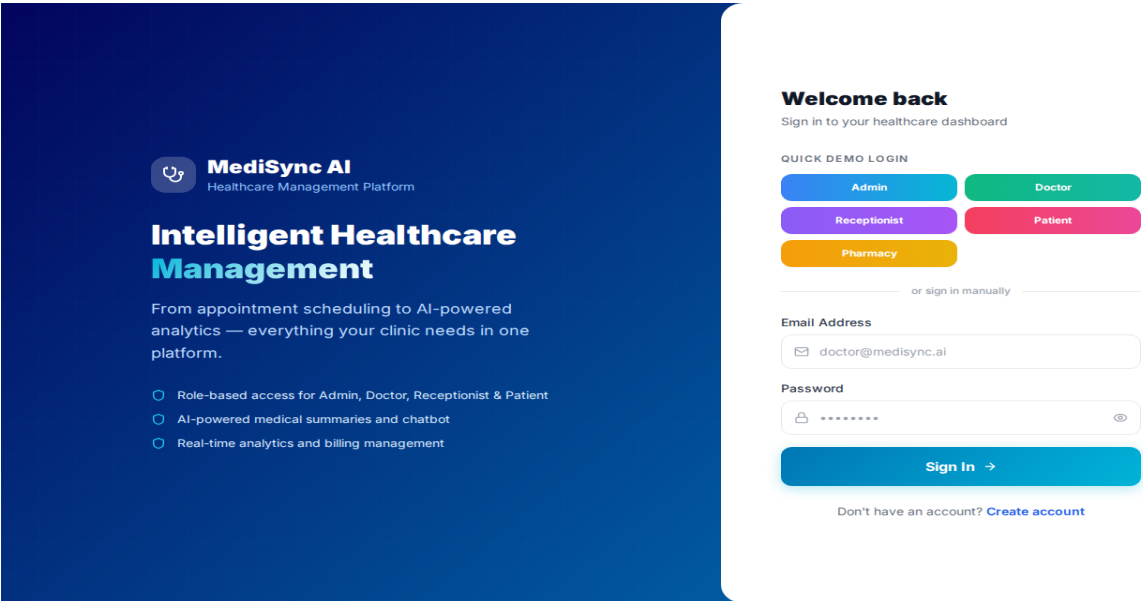


Figure 1: MediSync AI User Authentication GUI Page

Demo Login Credentials Table:

Role	Email Address	Password
Admin	admin@medisync.com	password123
Doctor	doctor@medisync.com	password123
Receptionist	receptionist@medisync.com	password123
Patient	patient@medisync.com	password123
Pharmacy	pharmacy@medisync.com	password123

3. Admin Portal Guide

The Admin Dashboard provides real-time oversight of hospital KPIs, patient directories, doctor scheduling, and financial revenues.

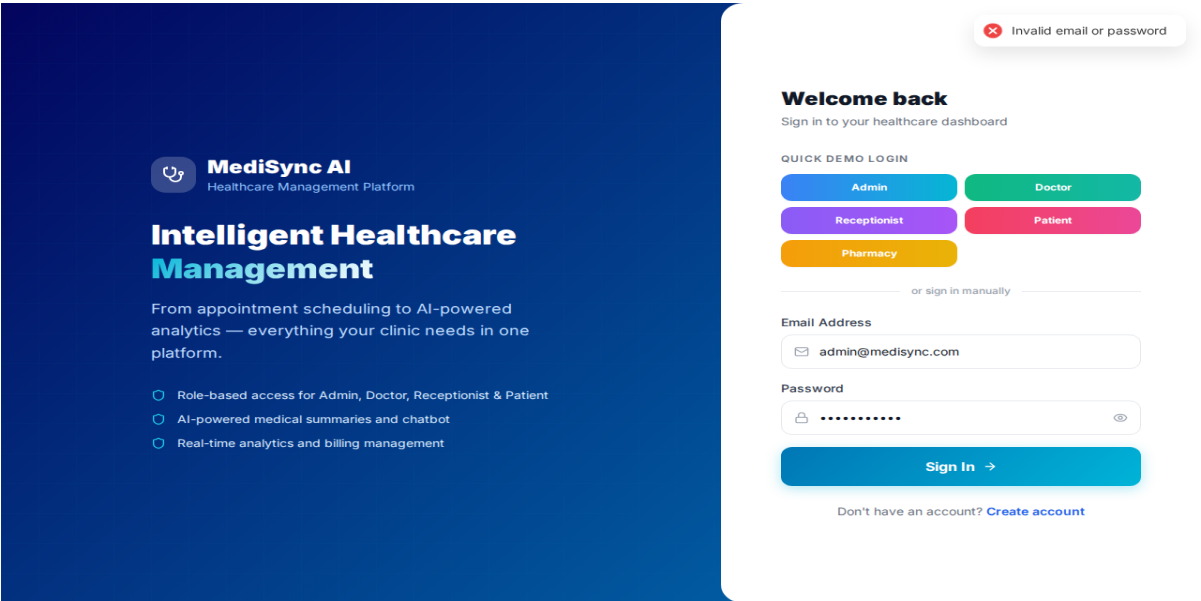


Figure 2: Admin Portal Executive Dashboard GUI

Step-by-Step Admin Navigation:

- 1. **Dashboard Overview:** Monitor live widgets for Total Patients, Active Doctors, Today's Appointments, Monthly Revenue, Pending Bills, and Emergency Alerts.
- 2. **Patient Directory** (`/dashboard/admin/patients`): Click **Patients** on the sidebar to search, edit, or onboard new patient records.
- 3. **Doctor Onboarding** (`/dashboard/admin/doctors`): Click **Doctors** to add medical staff, set consultation fees, shift schedules, and upload digital signatures.
- 4. **Billing Oversight** (`/dashboard/admin/billing`): Track itemized invoice generation and monitor payment collection metrics.

4. Doctor Portal Guide

Clinicians use this portal to manage patient appointments, view EHR medical histories, and issue signed e-prescriptions.

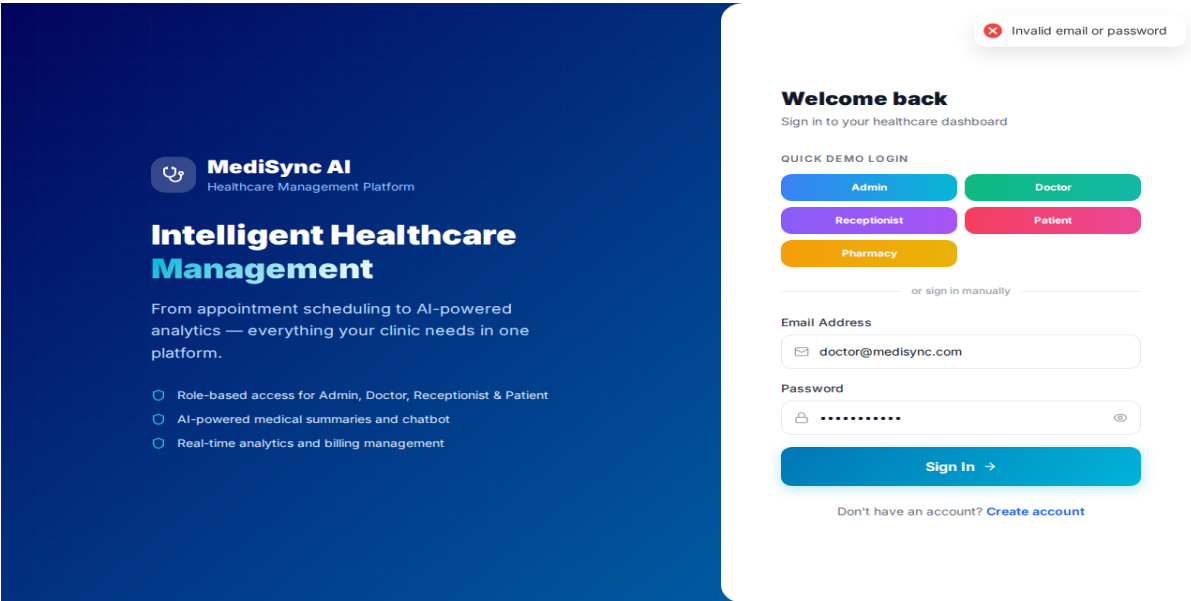


Figure 3: Doctor Consultation Dashboard GUI

Step-by-Step e-Prescribing Workflow:

- 1. Navigate to **Prescriptions** on the left sidebar navigation.
- 2. Click **New Prescription** at the top right of the screen.
- 3. Select patient name, add prescribed drugs with dosages (e.g. 500mg) and frequency (e.g. 1-0-1).
- 4. Save the prescription. The doctor's verified digital signature is automatically stamped.

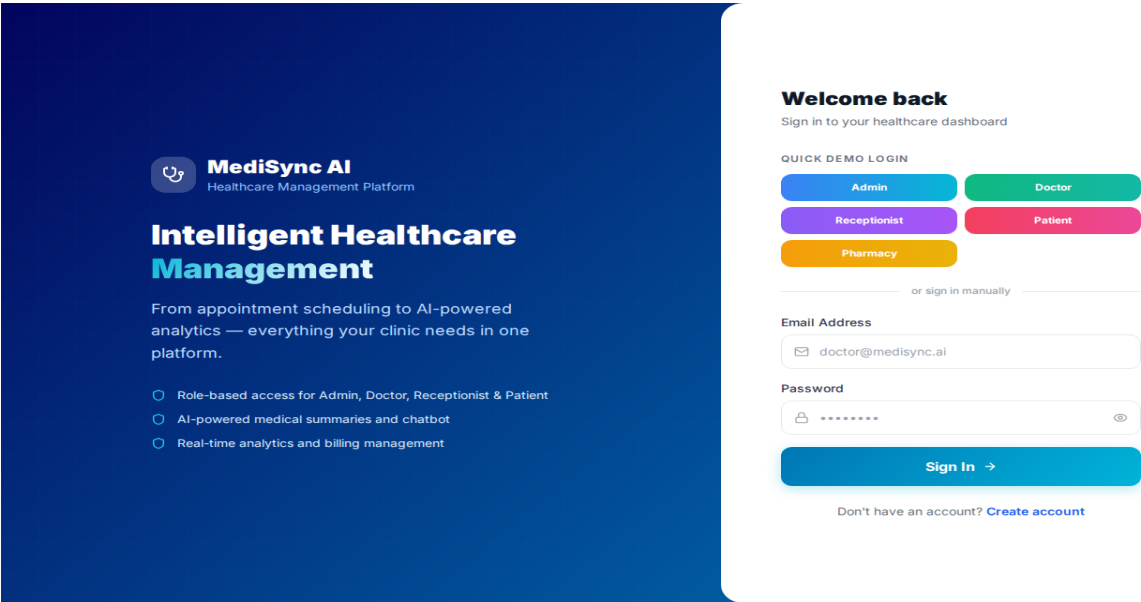


Figure 4: Doctor Electronic Prescribing Interface GUI

5. Receptionist Portal Guide

Designed for front-desk triage, walk-in registration, digital token generation, and consultation queue optimization.

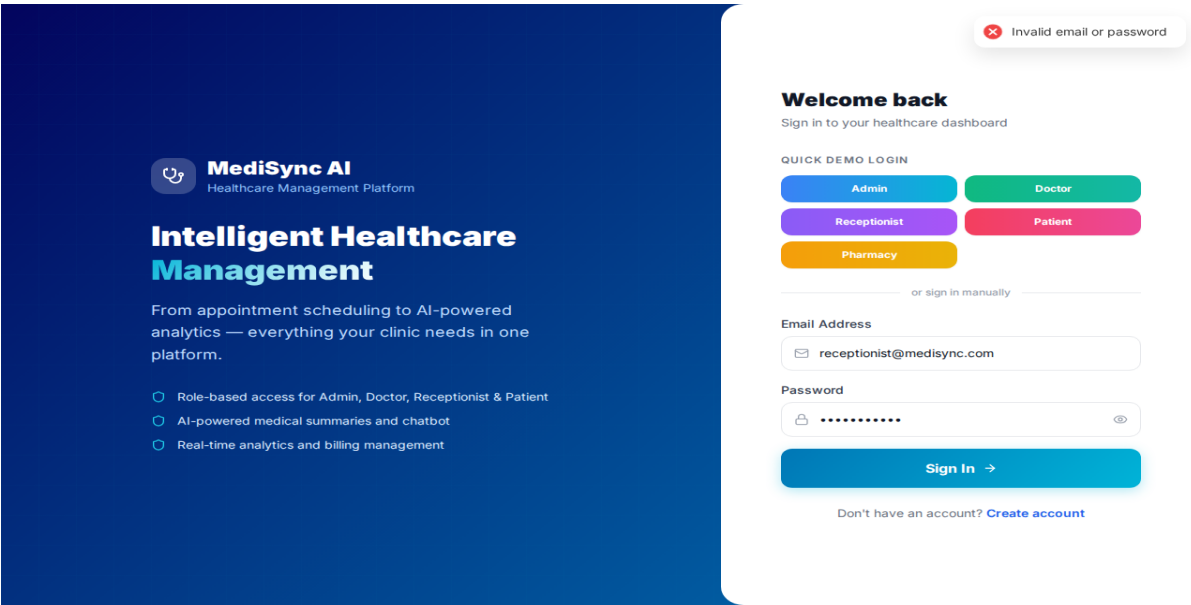


Figure 5: Receptionist Overview Dashboard GUI

Step-by-Step Queue Management:

- 1. Click **Queue Management** in the sidebar menu.
- 2. View active waiting list categorized by department and doctor availability.
- 3. Issue new sequential tokens or set priority to **Emergency** for urgent patient arrivals.

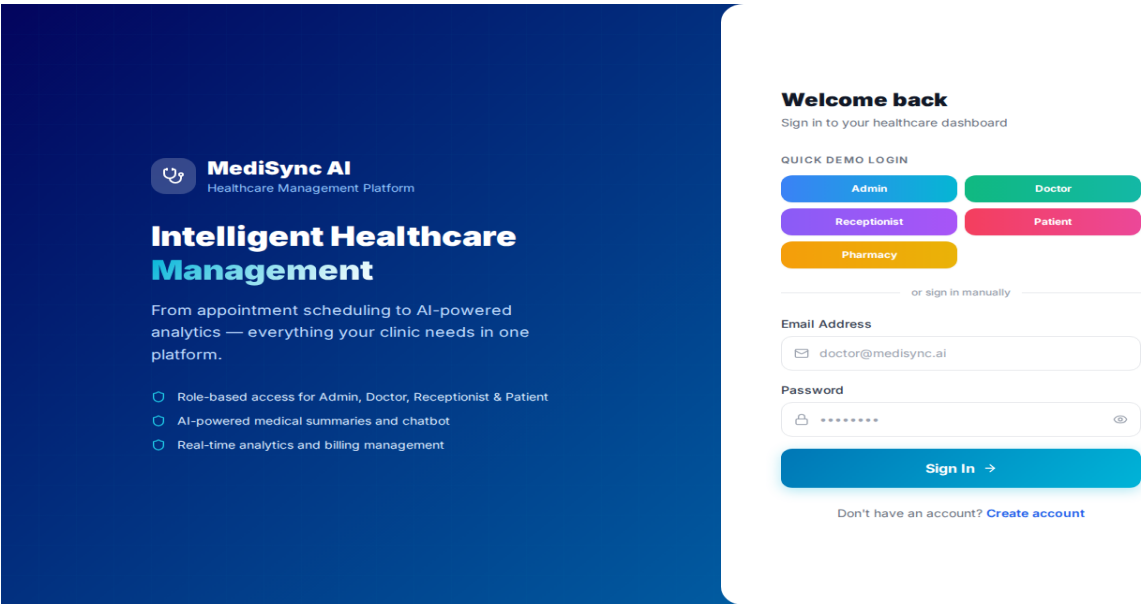


Figure 6: Front Desk Queue & Token Management GUI

6. Patient Portal Guide

Empowers patients to manage their health journey with online booking, medical history access, and bill payments.

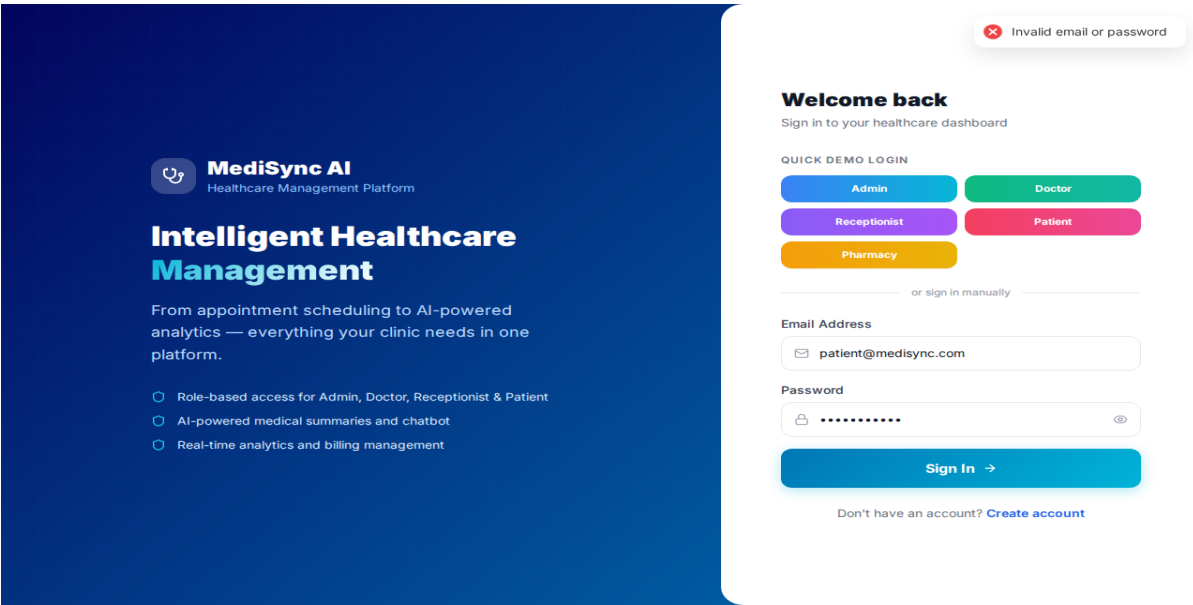


Figure 7: Patient Self-Service Dashboard GUI

Step-by-Step Patient Actions:

- 1. **Book Consultation:** Click **Book Appointment**, choose department, physician, date, and preferred time slot.
- 2. **View Medical Vault:** Click **Medical History** to review diagnosis summaries and download signed prescriptions.
- 3. **Pay Bills Online:** Click **My Bills** to view breakdown and process digital payment.

7. Pharmacy Portal Guide

Connects doctors with pharmacists for real-time prescription verification and drug dispensing.

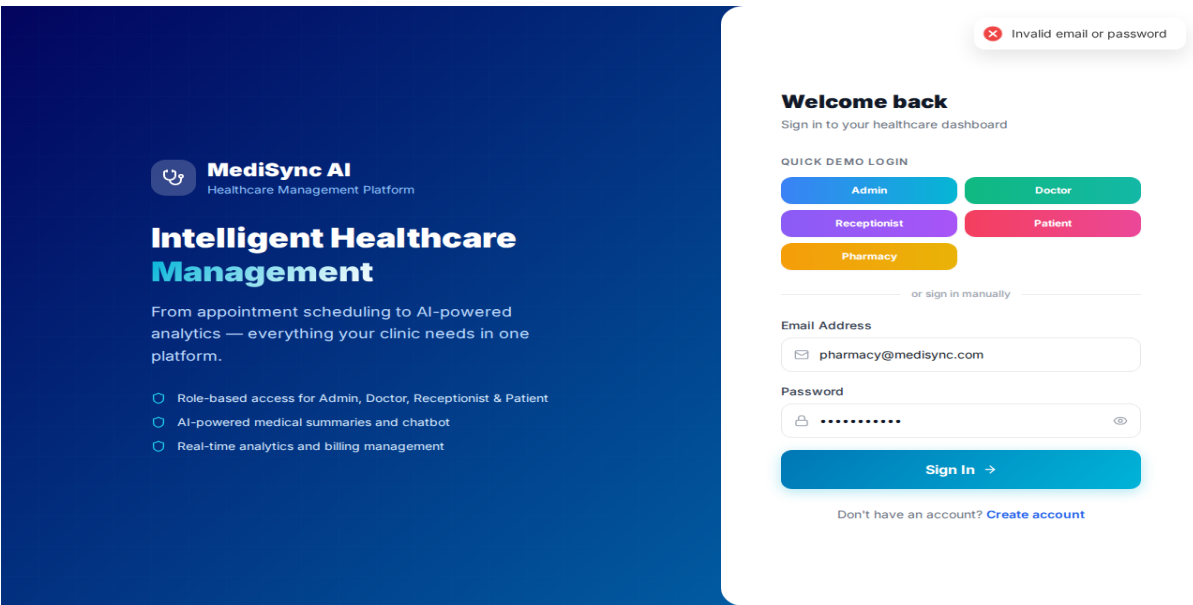


Figure 8: Pharmacy Prescriptions Queue & Dispensing GUI

Step-by-Step Dispensing Workflow:

- 1. View live incoming electronic prescriptions.
- 2. Verify medication dosage, administration frequency, and physician signature.
- 3. Click **Mark as Dispensed** once the medicine has been provided to the patient.

8. Shared AI & Emergency Tools

MediSync AI Clinical Assistant:

Enter patient symptoms to generate intelligent differential diagnosis insights and formatted summaries.

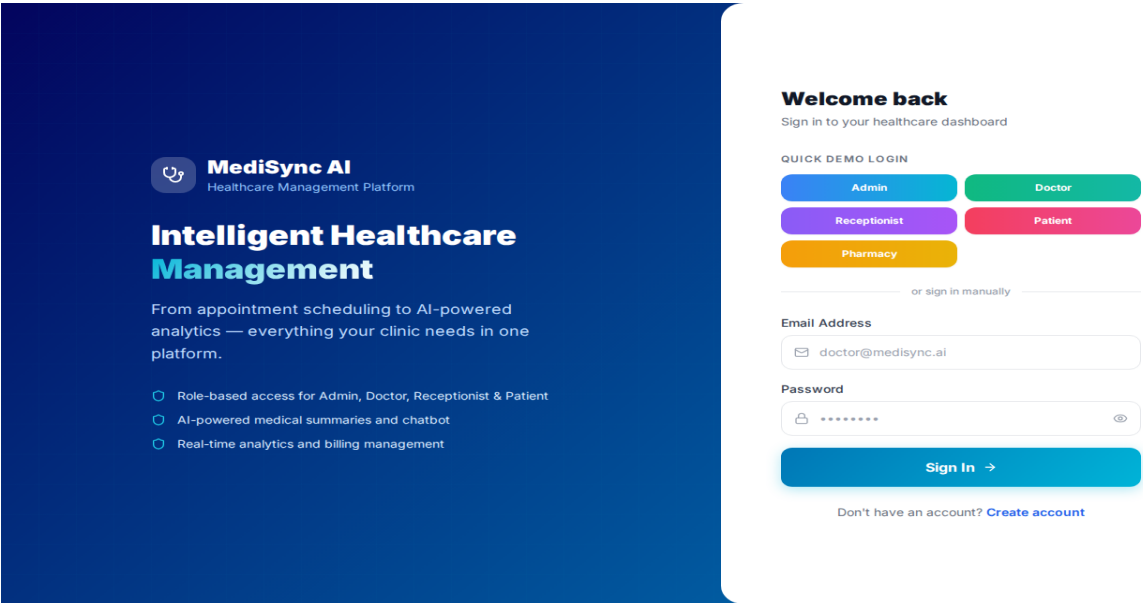


Figure 9: MediSync AI Assistant GUI

Emergency Medical Wallet:

Provides immediate access to critical emergency data: blood group, severe allergies, and emergency contact details.

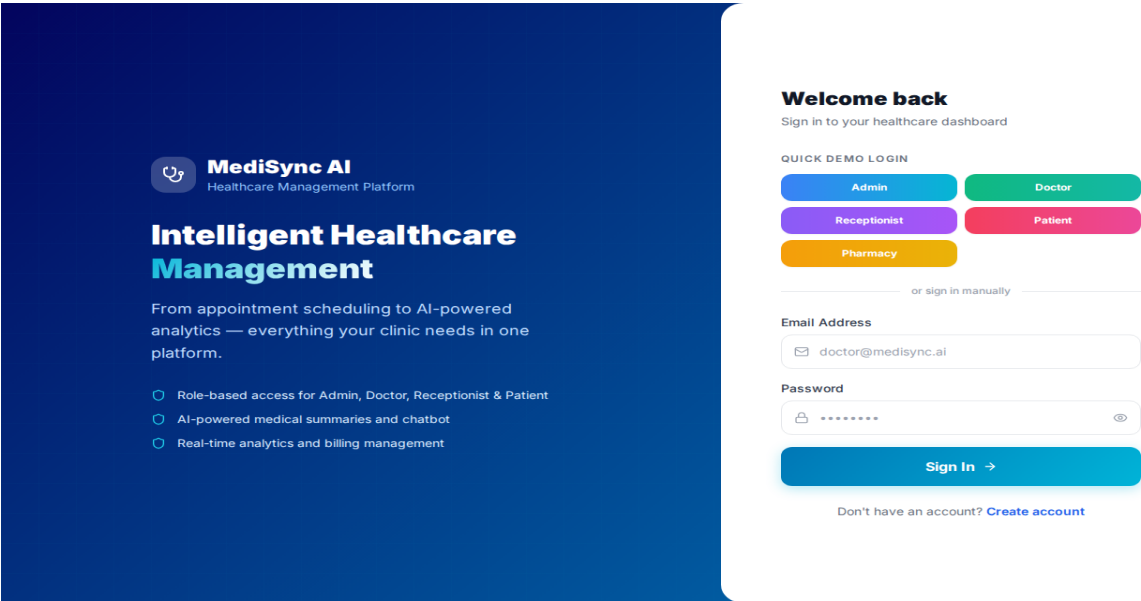


Figure 10: Emergency Medical Wallet GUI

9. API Specifications & Integration

REST API endpoints communicate using JSON format and require JWT Bearer authentication headers.

Endpoint	Method	Description
/api/auth/login	POST	User authentication & JWT generation
/api/patients	GET / POST	Fetch patients list or register new patient
/api/doctors	GET / POST	Fetch doctors directory or add doctor
/api/appointments	GET / POST	Fetch appointments or schedule new session
/api/prescriptions	GET / POST	Fetch prescriptions or generate new e-prescription
/api/invoices/:id/pay	POST	Process digital payment for invoice

10. Troubleshooting & FAQ

Q: How are emergency appointments prioritized?

A: Setting appointment priority to **Emergency** immediately places the patient at the top of the doctor queue and broadcasts system-wide notifications.

Q: Where are digital signatures attached?

A: Doctors upload their digital signature in Profile Settings. The system stamps this signature automatically onto generated e-prescriptions.